

The University of Lahore
Department of Computer Science & IT
Final Term Examination



Fall-2025

Course Title:	Web Engineering	Course Code:	CS02407	Credit Hours:	3
Course Instructors:	Mr. M. Zaman Aslam	Program Name:	BSCS		
Time Allowed:	120 Minutes	Maximum Marks:	40		
Date:	30-01-2026	Course Mentor:	Mr. Abdul Ghaffar		
Student's Name:		Signature:			
Reg. No:		Section:	6B		

Important Instructions / Guidelines:

- Attempt objective part on the question paper before moving to the subjective part.
- Solve subjective part on the answer sheet provided separately.
- Cutting, overwriting, using correction fluid is not allowed in the objective part.
- Your paper will be marked as ZERO, if involved in any kind of cheating.

Objective Part

Time: 10 Minutes

Question # 1: Circle the correct option.

(5 Marks)

1.	Which PHP superglobal is used to collect data sent through URL parameters? a. \$_POST b. \$_GET c. \$_REQUEST d. \$_URL
2.	Where are validation rules usually written? a. Model b. Route c. View d. Controller
3.	How can you add a comment in a JavaScript file? a. * This is a comment *\ b. <!-- This is a comment --> c. // This is a comment d. /* This is a comment */
4.	Which PHP feature allows data to persist across multiple pages without using a database? a. Cookies b. Super Global Variables c. Sessions d. Variables

5.	Which Artisan command is commonly used to create a controller? a. php artisan make:model controller b. php artisan make:controller c. php artisan create:model controller d. php artisan create:controller
6.	What is the result of the expression 5 + "10 apples" in PHP? a. 15 b. error c. 510 d. 5
7.	Which method is used to insert data using Eloquent? a. Insert() b. store() c. Add() d. save()
8.	In JavaScript, which method is used to add a new element to the end of an array? a. push() b. append() c. Unshift() d. Shift()
9.	In MVC View depends on? a. Models b. Controllers c. Templates d. None
10.	Which function is best suited to prevent SQL injection when using MySQLi? a. mysqli_query() b. mysqli_escape_string() c. mysqli_real_escape_string() d. mysqli_query_escape_string()

The University of Lahore
Department of Computer Science & IT
Final Examination



Fall-2025

CS02407-Web Engineering

Subjective Part

Time: 110 Minutes

Total Marks: 35

Question # 2: Answer following Short Questions.

(10+5 Marks)

Part 1

- i. A login form sends user data to the server securely. Which PHP super global should be used to receive this data?
- ii. Describe the difference between GET and POST methods in form submission.
- iii. A website needs to maintain user login information across multiple pages. Explain how sessions are used in PHP.
- iv. Explain the difference between `print_r()` and `var_dump()` in PHP. Why we use these?
- v. User input from a form is received as a string, but arithmetic operations are required. Explain how JavaScript handles string to number conversion.

Part 2

- a) A form submits user data that must be checked before inserting into the database. Explain how validation is implemented in Laravel.
- b) A developer wants to organize database tables and manage changes over time. Explain the purpose of migrations in Laravel.

Question # 3: Read and Solve the Scenario given below.

(20 Marks)

Note: Carefully read the whole scenario first and then perform the following operations as mentioned below. We did different assignments and practice during the lab. This is not a complete scenario so write code of only demanded pages you will get puzzled if you do not follow the instructions.

The University of Lahore plans to develop an Online Course Review Management System to allow students to share their learning experience and help improve teaching quality. The system will be accessible only to registered students and will provide a secure and user-friendly interface. The system allows students to:

- I. Register and log in securely
- II. View a list of available courses

III. Submit a review for a selected course

IV. Edit their previous submitted reviews

The system must be developed using HTML, PHP, and MySQL only.

1. **Database Design:** Design a MySQL database named **course_review_system**.

Create an ER diagram only (no SQL code) to represent the following entities and their relationships:

- I. students (id, name, email, password)
II. courses (id, course_title, instructor_name)

- III. reviews (id, student_id, course_id, rating, review_text, review_date)

Note: (ER diagram drawn using rectangles for entities, ovals for attributes, and diamonds for relationships)

2. **User Authentication:** Develop the following components:

- HTML forms for student registration and login
- PHP scripts using MySQLi procedural to:
 - Validate login credentials
 - Prevent duplicate email registration
- Create registration page with fields you mentioned in ER diagram.

3. **Submit Course Review :** After successful login, a student selects a course from a dropdown menu and submits a review consisting of:

I. Rating (1–5)

II. Review text (textarea)

Write PHP code to:

- Insert the review into the database
- Ensure that a student can submit only one review per course
- Display a success or error message accordingly

4. **Update Review:** Create a “My Reviews” page that:

- Displays all reviews submitted by the logged-in student
- Shows an “Edit” button next to each review
- Opens a pre-filled form allowing the student to update:
 - Rating
 - Review text

The PHP script should:

- Fetch review details using the review ID
- Execute an UPDATE query

****Good Luck****